Enhancing Collaboration and Mitigating Conflict Between Subtasks for Remote Sensing Semantic Change Detection

Zhihao Li, Yikun Liu, Minghao Liu, and Gongping Yang

***Abstract*— Semantic change detection (SCD) consists of two separate but not independent parts, that is, binary change detection (BCD) and semantic segmentation (SS), which aims to simultaneously locate changed areas and provide their semantic categories in bi-temporal remote sensing (RS) images. Although the existing SCD methods have achieved excellent performance, they still have two challenges: 1) insufficient collaboration between BCD and SS: they cannot efficiently leverage semantic information provided by SS to further improve the performance of BCD and 2) existing conflict between BCD and SS: they usually share input between SS and BCD branches to benefit from joint optimization. However, the shared input will be constrained by the task orientations of BCD and SS, which may produce conflict. Specifically, the task orientation of BCD aims to align the distribution of bi-temporal domains, which inevitably results in the constraint of the shared input mining domain-specific features and further causes degradation of SS performance. Similarly, the task orientation of SS also compromises the domain alignment process of BCD as the shared input mines domain-specific features guided by the task orientation of SS for improving its performance. Therefore, we propose the enhancing collaboration-mitigating conflict network (EC-MCNet). First, a semantic enhancement module (SEM) is designed to enhance interclass variation and intraclass similarity of semantic features, which are beneficial for SS and BCD. Second, we no longer share input but share semantic contents between the SS and BCD branches to mitigate the conflict. Specifically, we build a domain style removal module (DSRM), which ensures that the input of the BCD branch is removed from domain-specific styles and has the same semantic contents as the inputs of SS branches. In this way, the conflict between BCD and SS can be mitigated and the advantage of joint optimization can be preserved. Third, we design a difference enhancement module (DEM) to enhance collaboration between SS and BCD, which leverages not only the attention of difference features but also the semantic similarity between bi-temporal features to enhance and identify changed areas of bi-temporal features. Extensive experimental results validate that our method outperforms state-of-the-art (SOTA) performances on two benchmark datasets for the SCD. The source code is available at https://github.com/yihui1230/ECMCNet**

***Index Terms*— Enhancing collaboration, mitigating conflict, remote sensing (RS), semantic change detection (SCD).**

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I. Introduction

As MORE semantic change detection (SCD) datasets such as SECOND [1] and Landsat-SCD [2] become available, SCD is getting more attention. Compared to binary change detection (BCD), SCD aims to simultaneously locate changed areas and provide their semantic categories in bi-temporal remote sensing (RS) images. Therefore, SCD is more effective and crucial for analyzing urban changes, managing resources, monitoring pasture changes, assessing agricultural damage, warning disasters, and so on.

Previous works have given SCD a lot of attention. Xian et al. [3] first identified changed areas based on the change vectors and then classified their semantic categories. Bruzzone and Serpico [4] and [5] leveraged the Bayes rule to iteratively estimate the semantic changes. However, the previous works have deficiencies in modeling complicated semantic information, especially in the face of complex RS scenarios.

With the development of deep learning, a large number of neural network-based methods have been proposed for RS tasks, whose learnable paradigm ensures that there is no deviation from their task orientations. Convolutional neural networks (CNNs) are capable of extracting abstract features and local details. Thus, [6], [7], [8] adopted a Siamese weight-shared CNN-based encoder to extract bi-temporal deep features from bi-temporal images and then obtained difference features. To expand the receptive field, [9] adopted a Siamese weight-shared Transformer-based encoder to extract bi-temporal global features, which can model global relationships but is deficient in extracting local details compared with the CNN.

Specific to SCD, [10] and [11] leveraged the advantages in extracting abstract features of the CNN to execute the SCD. Yuan et al. [2] adopted a Transformer as an encoder to capture long-range spatial–temporal features, whose global modeling capability facilitated the extraction of contextual semantic information. Ding et al. [12] proposed a novel CNN–Transformer hybrid network, which investigated jointly the spatial–temporal dependencies based on global relationship modeling and extracted fine-grained details.

From the perspective of structural design, the existing SCD models can be divided into three parts: single-branch structure, dual-branch structure, and triple-branch structure. The single-branch structure [2] only adopts one difference

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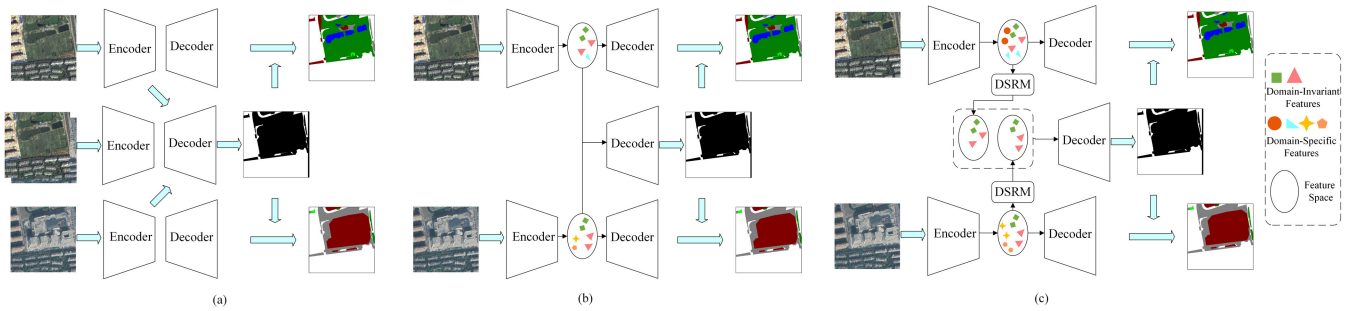


Fig. 1. Comparison of our structure design with two existing popular structure designs. (a) Structure design of the triple encoder, whose inputs of decoders are independent. (b) Structure design of two encoders, whose inputs of decoders are shared. (c) Our structure design of two encoders, whose inputs of decoders are different but share the same semantic contents. The inputs of SS branches can retain domain-specific features for SS and the inputs of the BCD branch can mitigate domain-specific features for BCD, which mitigates the conflict between BCD and SS.

branch to obtain the SCD map. Although the single-branch structure has fewer parameters compared with other structures, it generally needs to identify "from–to" semantic category transitions. Specifically, if the types of semantic categories are N in bi-temporal images, the model needs to totally consider $N(N-1)$ types of changed semantic category combinations, which makes accurate predictions extremely difficult. The dual-branch structure [13] usually sets two semantic segmentation (SS) branches and unchanged areas are considered as one of the categories, which can directly obtain SCD maps. However, the SCD maps of the dual-branch structure still consider the change information, which indicates that interaction between the two SS branches is indispensable. Therefore, the dual-branch structure may lead to imbalance and interference between BCD and SS. Differently, [11], [14], [15], [16], [12], and [17] adopt a triple-branch structure, which comprises one BCD branch and two SS branches. It leverages BCD maps as a mask and SS maps to generate SCD maps. The structure effectively mitigates interference and imbalance between BCD and SS. As shown in Fig. 1, the triple-branch structure can be further divided into the schemes of Fig. 1(a) and (b) depending on whether their inputs of the decoder are shared or not. Compared to the scheme of Fig. 1(a), the scheme of Fig. 1(b) can better enhance and benefit from collaboration between BCD and SS. Li et al. [11] also indicated that the scheme of Fig. 1(b) is capable of extracting more representative features for both SS and BCD. Specifically, the scheme of Fig. 1(b) allows the shared input to better perceive and acquire semantic information. Instead, the scheme of Fig. 1(a) is limited by the semantic gap of the features extracted by different encoders. Therefore, more precise SCD maps will be outputted from Fig. 1(b) based on the semantic enhanced shared input.

Challenges: Although the triple-branch structure has greatly improved the performance of SCD, there are still two challenges in SCD: 1) insufficient collaboration between BCD and SS: [7] and the success of triple-branch structure both indicate that semantic information provided by SS can improve the performance of BCD. However, the existing SCD methods still identify the changed areas based on BCD ways and fail to further establish the relationship between the semantic information provided by SS and changed areas and 2) existing conflict between BCD and SS: although the scheme of shared input of triple-branch structure has greatly improved the performance of SCD, it introduces the conflict between BCD and SS. From the perspective of domain learning, BCD is a domain alignment task, whose task orientation aims to align the distribution of bi-temporal domains. In other words, the task orientation of BCD aims to mitigate domain-specific features and retain domain-invariant features of the shared input, while the loss of domain-specific features in the shared input inevitably leads to performance degradation of SS. Similarly, the task orientation of SS also compromises the domain alignment process of BCD as the task orientation of SS forces the shared input to mine domain-specific features for improving SS performance. As shown in Fig. 1(b), the scheme causes the shared input to lose part domain-specific features for SS and retains part domain-specific features for BCD, which both indicate that the scheme is suboptimal.

To address the above challenges, we propose the enhancing collaboration-mitigating conflict network (EC-MCNet). First, we design a semantic enhancement module (SEM), which groups features and processes them separately. In this way, the features that express the class-specific semantic information contained in each group can be generated. Therefore, interclass variation and intraclass similarity of semantic features can be enhanced, which are beneficial for SS and BCD. Second, we no longer share input but share semantic contents between SS and BCD branches, which not only mitigates the conflict but also preserves the advantage of the scheme of shared input. Specifically, we build a domain style removal module (DSRM), which removes domain-specific styles of SS branch inputs by processing their low-frequency to generate the BCD branch input. Notably, it only processes their low-frequency that is related to styles, which ensures the input of the BCD branch has the same semantic contents as the inputs of SS branches. Furthermore, the input of the BCD branch can still perceive and benefit from semantic information provided by SS based on the shared semantic contents. In this way, the inputs of SS branches are relieved of the constraint of mining domain-specific features and bi-temporal domains can be aligned more efficiently. As shown in Fig. 1(c), the inputs of SS branches can retain domain-specific features for SS and the input of the BCD branch can mitigate domain-specific features for BCD. Third, we design a difference enhancement module (DEM) to enhance collaboration between SS and BCD. Liu et al. [18] indicated that each channel usually has

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different sensitivities on different classes, which can correspond to specific semantics. Thus, we argue that the effect of semantic information provided by SS can be described as the bi-temporal features having more class-specific representations that are more organized in specific channels. Furthermore, it may be feasible to enhance and identify changed areas based on the representations similarity of bi-temporal features on corresponding channels and across channels. Therefore, DEM leverages not only the attention of difference features but also the semantic similarity between bi-temporal features to enhance and identify changed areas, which further establishes the relationship between the semantic information and changed areas in SCD.

The major contributions of our work are as follows.

1) We propose the EC-MCNet, which effectively enhances collaboration and mitigates conflict between BCD and SS.
2) We design the SEM to enhance interclass variation and intraclass similarity of bi-temporal semantic features, which are beneficial to BCD and SS.
3) We point out the conflict of the existing triple-branch structure from the perspective of domain learning. Therefore, we no longer share input but share semantic contents between SS and BCD branches based on our DSRM, which not only mitigates the conflict but also preserves the advantage of the scheme of shared input. In this way, the inputs of SS branches are relieved of the constraint of mining domain-specific features and bi-temporal domains can be aligned more efficiently.
4) To enhance collaboration between SS and BCD, we design DEM that can establish the relationship between the semantic similarity and changed areas, which makes effective use of semantic information provided by SS to enhance and identify changed areas.

The rest of this article is organized as follows. Section II gives a brief review of the most related work. In Section III, we present the details of the proposed method. Section IV reports experimental results and corresponding analyses. Finally, Section V summarizes this article.

II. Related Work

A. Semantic Segmentation

SS is a fundamental task in RS image processing, whose goal is to categorize each pixel of RS images into a class. Therefore, SS has great significance in road prospecting, urban planning, and disaster prevention. With the deep learning networks that have been extensively applied in the field of SS, SS achieved great progress. Chen [19], [20], [21] proposed the DeepLab series of networks, which leveraged their atrous space pyramid pooling (ASPP) to focus on multiscale features. Moreover, ASPP increased the receptive field without reducing the image size and thus improved the accuracy of SS maps. Zhao et al. [22] proposed the pyramid scene parsing network, whose pyramid pooling module was capable of capturing multiscale features and fusing them. Zheng et al. [23] indicated that each channel of a feature map represented a specialized detector, thus it proposed to capture the channel attention with various scales for achieving dynamic and adaptive feature fusion in a contextual scale-aware manner. The effective usage and fusion of multiscale features significantly improve the accuracy of SS maps. For another thing, interclass similarity and intraclass variation of features are also challenges for SS. Yu et al. [24] proposed a discriminative feature network that contained a smooth network and border network to enhance interclass variation and intraclass similarity of the semantic features. Huang et al. [25] proposed three loss functions to encourage more compact class representations within each class and maximize the distance between different class centers.

B. Binary Change Detection

The goal of BCD is to detect and identify changed areas between bi-temporal images. Benefit from the rapid development of deep learning, a large number of neural network-based methods [6], [26], [27], [28], [29], [30] have been proposed for RS change detection (RSCD). For BCD, how to enhance intraclass similarity and interclass variation across bi-temporal domains is the most important challenge. From the perspective of domain learning, BCD is a domain alignment task because the task orientation of BCD aims to align intraclass representations across bi-temporal domains by removing domain-specific features. To further align them, a lot of work has been focused on domain style removal. For instance, [31] mitigated the impact of domain style shifts by embedding the mean and variance of the features. Ji et al. [32] leveraged maximum mean discrepancy as a constraint to align bi-temporal domains, which can reduce the distribution difference between the bi-temporal features. Liu et al. [33] filtered style relevance information in the frequency domain to eliminate the interference of image style. However, the operation was also imposed on the high frequencies, which inevitably compromised feature contents. On the other hand, the enhancement of the changed areas of the bi-temporal features is also important for BCD. Lei et al. [34], [35], [36] all leveraged the difference features obtained by subtraction of bi-temporal features to generate difference attentions, which were adopted to enhance the changed areas of the bi-temporal features.

C. Semantic Change Detection

SCD consists of two separate but not independent parts, that is, BCD and SS, which aim to simultaneously locate changed areas and provide their semantic categories in bi-temporal RS images. Ding et al. [15] proposed a semantic consistency loss function to align the semantic and change representations. Li et al. [11] focused on the design of the decoder, which better enhanced the capitalization of the feature interactions across the BCD and SS decoding heads from the perspective of multitask learning. Ding et al. [12] investigated the joint modeling of the spatial–temporal dependency to improve the learning of semantic changes. Chang et al. [16] proposed a new triple-branch network, which took the joint bi-temporal features as the unified input of decoders. In this way, each branch can perceive the bi-temporal semantic information without any additional interaction between them. The existing methods

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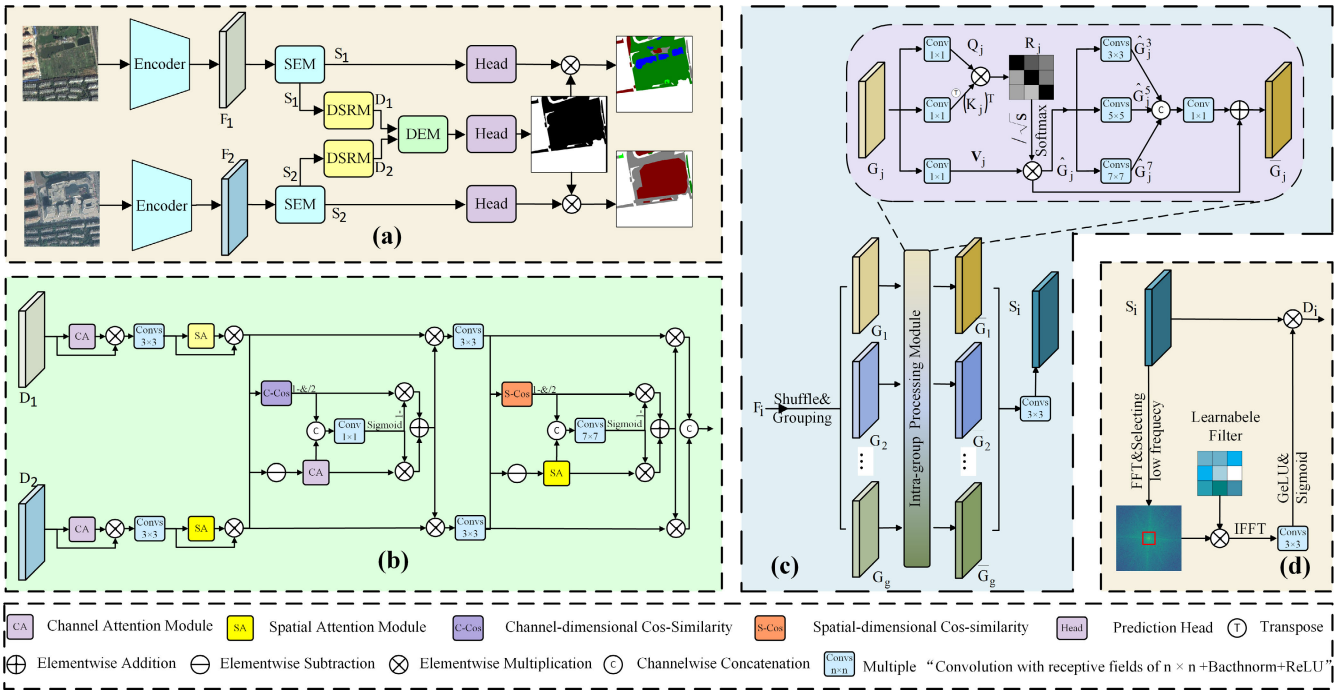


Fig. 2. Overall framework and detail of the proposed EC-MCNet. The bi-temporal images are first extracted by the shared weighted encoder. Next, the outputs of the encoder F_1 and F_2 are fed into SEM to obtain S_1 and S_2, which are enhanced with interclass variation and intraclass similarity. Compared to existing methods, we leverage S_1 and S_2 to generate SS maps but take D_1 and D_2 as inputs of BCD branch that are removed from domain-specific styles by DSRM, which can mitigate conflicts between BCD and SS. Then, D_1 and D_2 pass through DEM to enhance their changed areas by leveraging semantic similarity, which can efficiently enhance collaboration between BCD and SS. After that, the output of DEM turns into the final CD maps. Finally, the BCD maps mask the SS maps to generate SCD maps. (a) Overall framework of the proposed EC-MCNet. (b) DEM. (c) SEM. (d) DSRM.

have achieved impressive performances in SCD. However, Iizuka et al. [37] proposed a data augmentation method for domain generalization in SS, whose results suggested the domain generalization process comprised the performance of the source domain. Therefore, we argue that BCD as a domain alignment task inevitably comprises the performance of SS. How to mitigate the conflict between BCD and SS is a challenge for SCD. On the other hand, the existing SCD methods enhanced the changed areas of the bi-temporal features by attention to the difference features as BCD, which suggested the semantic information provided by SS was not being used effectively. Liu et al. [18] indicated that each channel usually has different sensitivities on different classes, which can correspond to specific semantics. Therefore, we argue that the effect of semantic information provided by SS can be described as the bi-temporal features having more class-specific representations that are more organized in specific channels. How to effectively leverage the prior knowledge that class-specific representations are more organized in specific channels to enhance and identify changed areas of the bi-temporal features is another challenge for SCD.

III. Proposed Method

In this section, we first present an overview of the proposed EC-MCNet. Then, we give the details of the proposed SEM, DSRM, and DEM.

A. Overview

As illustrated in Fig. 2(a), given two bi-temporal images I_1 and $I_2 \in \mathbb{R}^{3\times H\times W}$, which are first extracted by the shared weighted encoder to output F_1 and $F_2 \in \mathbb{R}^{C\times H/4\times W/4}$, respectively. Regarding our encoder, we adopt the encoder of the TED in [12], whose outputs are fine-grained and have a more precise representation of spatial information. Therefore, we only use three lightweight prediction heads for decoding and predicting the corresponding branches, which consist of convolutions with receptive fields of 3×3, batch normalization layers, and the ReLU activation function. Next, F_1 and F_2 pass through the SEM to obtain S_1 and $S_2 \in \mathbb{R}^{C\times H/4\times W/4}$, which are enhanced with interclass variation and intraclass similarity. Then, S_1 and S_2 are fed into the SS prediction head to generate SS maps. Meanwhile, DSRM generates D_1 and $D_2 \in \mathbb{R}^{C\times H/4\times W/4}$ as the input of the BCD branch by removing the domain-specific styles of S_1 and S_2, respectively. In this way, D_i without the domain-specific styles has the same semantic contents as S_i, which mitigates the conflict between BCD and SS. After that, D_1 and D_2 are fed into the DEM to enhance their changed areas based on the semantic similarity, which can efficiently enhance the collaboration between BCD and SS. Finally, the BCD maps outputted by the BCD prediction head mask the SS maps to generate SCD maps.

B. Semantic Enhancement Module

Recently, interclass similarity and intraclass variation have always been challenges for SS, which cause decision boundaries between classes to blur and thus compromise the performance of SS. On the other hand, the core of BCD is to enhance interclass variation and intraclass similarity across bi-temporal domains. Therefore, enhancing interclass variation

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and intraclass similarity is crucial for SCD. Furthermore, we group features and process them separately. In this way, the features that express the class-specific semantic information contained in each group can be generated, which can efficiently enhance interclass variation and intraclass similarity. Technically, we first shuffle them randomly and re-group them into $g \in \mathbb{Z}$ different groups, the feature of each group has $d \in \mathbb{Z}$ channels. Notably, the product of d and g is equal to C. Then, the intragroup processing module mines the class-specific semantic features of each group. Specifically, the module first models the global relationship between channels within each group, the channel relationship allows each channel to acquire information concerning similarities or differences with other channels, thus leading better to mining class-specific semantic features. For feature G_j in group j, we generate query (Q_j), key (K_j), and value (V_j) by three convolution operations

$$Q_j = G_j W^{q_j}, \quad K_j = G_j W^{k_j}, \quad V_j = G_j W^{v_j} \tag{1}$$

where W^{q_j}, W^{k_j}, and $W^{v_j} \in \mathbb{R}^{d\times d}$ are convolution operations. Then, we calculate the channel relationship $R_j \in \mathbb{R}^{d\times d}$ by a dot product operation on Q_j and $(K_j)^{\mathrm{T}}$, which further does the dot product of V_j to generate channel correlation enhanced $\hat{G}_j$

$$R_j = Q_j\left(K_j\right)^{\mathrm{T}}, \quad \hat{G}_j = \mathrm{Softmax}\left(\frac{R_j}{\sqrt{s}}\right)V_j \tag{2}$$

where s is a scale factor. Considering the variety of shapes and sizes of objects on RS images, we leverage multiple convolutions with different receptive fields to mine more class-specific clues of each group

$$\begin{aligned}
\hat{G}_j^3 &= \mathrm{Convs}_{3\times3}\left(\hat{G}_j\right)\\
\hat{G}_j^5 &= \mathrm{Convs}_{5\times5}\left(\hat{G}_j\right)\\
\hat{G}_j^7 &= \mathrm{Convs}_{7\times7}\left(\hat{G}_j\right)\\
\overline{G_j} &= \mathrm{Conv}_{1\times1}\left(\mathrm{Cat}\left(\hat{G}_j^3, \hat{G}_j^5, \hat{G}_j^7\right)\right) + \hat{G}_j
\end{aligned} \tag{3}$$

where $\mathrm{Cat}(\cdot)$ represents a channelwise concatenation operation, $\mathrm{Convs}_{n\times n}(\cdot)$ denotes multiple combinations of a convolution with $n \times n$ receptive field, a batch normalization layer, and a ReLU activation function. Finally, we gather $\overline{G_j}$ from each group and merge them by convolution operation. Compared to the input F_i, the output S_i has clearer class-specific features, which are efficiently enhanced with interclass variation and intraclass similarity.

C. Domain Style Removal Module

Although the shared input scheme of the triple-branch structure assists in enhancing the ability of the BCD branch to perceive semantic information, it introduces the conflict between BCD and SS. To retain the advantage of the shared input scheme as well as mitigate the conflict, we no longer share input but share the same semantic contents between BCD and SS branches. Specifically, we design DSRM to generate D_i by removing domain-specific styles of S_i, which replaces S_i as input of the BCD branch. On the other hand, the inputs of the SS branches are still S_1 and S_2. In this way, D_i has the same semantic contents as S_i, which ensures the advantage of the shared input scheme can be retained. Meanwhile, the conflict can be mitigated.

The technical details of our DSRM are described next. Inspired by the works on domain generalization in the frequency domain [37], [38], we are aware that the features in the frequency domain enable more precise decoupling of style and content. Therefore, we first map the input of DSRM S_i into frequency domain by fast Fourier transform (FFT) [39]

$$\mathcal{F}(S_i)(u, v) = \frac{1}{\sqrt{HW}}\sum_{h=0}^{H-1}\sum_{w=0}^{W-1} S_i(h, w)e^{-j2\pi\left(\frac{h}{H}u+\frac{w}{W}v\right)}. \tag{4}$$

Intuitively, the low-frequency information is related to the style of the image and high-frequency information is relevant to the contents of the image. Therefore, we only select the low frequencies of $\mathcal{F}(S_i)$ and process them by doing the dot product of a learnable filter, which aims to remove domain-specific style through the learnable way. Then, we map the processed $\mathcal{F}(S_i)$ into spatial domain by inverse FFT (IFFT) [39] to obtain S_i^p. Instead of using it directly, we leverage it as an attention map to remove the domain-specific style of S_i. The reason is that processing in the frequency domain may cause features not to be smooth when mapped to the spatial domain. Thus, we leverage S_i^p to generate an attention map in the spatial domain based on convolution operation with a Gaussian error linear unit (GeLU) and a sigmoid activation function. The way is also able to remove the domain-specific style and can further ensure that the output features are smooth. The detailed operations are as follows:

$$D_i = \mathrm{Sigmoid}\left(\mathrm{GeLU}\left(\mathrm{Convs}_{3\times3}\left(S_i^p\right)\right)\right) \cdot S_i. \tag{5}$$

Benefit from DSRM, we can share semantic contents between SS and BCD branches, which ensures the inputs of SS branches are relieved of the constraint of mining domain-specific features and bi-temporal domains can be aligned more efficiently. In summary, the advantage of a shared input scheme can be retained and the conflict can be mitigated.

D. Difference Enhancement Module

Enhancing changed areas of the bi-temporal features has always been crucial for BCD and SCD. However, the existing SCD methods still adopt the difference enhancement approach of BCD, which suggests that semantic information provided by SS is not fully utilized and further causes the collaboration between BCD and SS to be insufficient. Liu et al. [18] showed that each channel usually has different sensitivities on different classes, which can correspond to specific semantics. Therefore, we argue that the effect of semantic information provided by SS can be described as the bi-temporal features having more class-specific representations that are more organized in specific channels. To utilize the particular semantic information concerning class-specific representations distributed in specific channels, we propose to enhance and identify changed areas based on the similarity of the bi-temporal feature representations on corresponding channels and across channels. Therefore, the DEM is designed to leverage not

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only the attention of difference features but also the semantic similarity between bi-temporal features, which can efficiently establish the relationship between the semantic similarity and changed areas in SCD.

Technically, we first remove the effects of the irrelevant background of D_i by the channel attention module (CA) and the spatial attention module (SA) [40]

$$\begin{aligned} D_i^1 &= \text{Convs}_{3\times3}(\text{CA}(D_i)\cdot D_i) \\ D_i^2 &= \text{SA}\left(D_i^1\right)\cdot D_i^1. \end{aligned} \tag{6}$$

Then, we calculate the channel-dimensional semantic cos-similarity map CSM on the corresponding channel of D_i^2 and convert it to the inverse channel similarity map ICSM $\in [0,1]^{C\times1\times1}$

$$\begin{aligned} \text{CSM} &= C - \text{Cos}\left(D_1^2, D_2^2\right) \\ \text{ICSM} &= (1-\text{CSM})/2 \end{aligned} \tag{7}$$

where $C-\text{Cos}$ represents channel-dimensional semantic cos-similarity operation. Next, we acquire the channel attention map of difference features DCM $\in [0,1]^{C\times1\times1}$

$$\text{DCM} = \text{CA}\left(\text{abs}\left(D_1^2 - D_2^2\right)\right). \tag{8}$$

After that, we re-weight ICSM and DCM to obtain the final channel attention map FCM

$$\begin{aligned} &\text{weight_}C = \text{Sigmoid}(\text{Convs}_{1\times1}(\text{Cat}(\text{ICSM}, \text{DCM}))) \\ &\text{FCM} = \text{weight_}C\cdot\text{DCM} + (1-\text{weight_}C)\cdot\text{ICSM}. \end{aligned} \tag{9}$$

Furthermore, FCM is exerted on D_i^2 to enhance channels related to changed areas

$$D_i^c, = \text{Convs}_{3\times3}\left(\text{FCM}\cdot D_i^2\right). \tag{10}$$

On the other hand, the representation across channels of spatial pixels that belong to the same class should be similar in SCD. Therefore, we calculate spatial-dimensional semantic cos-similarity map SSM and convert it to the inverse spatial similarity map ISSM $\in [0,1]^{1\times H/4\times W/4}$

$$\begin{aligned} \text{SSM} &= S - \text{Cos}\left(D_1^c, D_2^c\right) \\ \text{ISSM} &= (1-\text{SSM})/2 \end{aligned} \tag{11}$$

where $S-\text{Cos}$ represents spatial-dimensional semantic cos-similarity operation. Similarity, we enhance the changed areas of D_i^c like the above channel enhancement strategy

$$\text{DSM} = \text{SA}\left(\text{abs}\left(D_1^c - D_2^c\right)\right) \tag{12}$$

$$\text{weight_}S = \text{Sigmoid}(\text{Convs}_{1\times1}(\text{Cat}(\text{ISSM}, \text{DSM})))$$

$$\text{FSM} = \text{weight_}S\cdot\text{DSM} + (1-\text{weight_}S)\cdot\text{ISSM} \tag{13}$$

$$D_i^s = \text{FSM}\cdot D_i^c. \tag{14}$$

Finally, we concatenate D_1^s and D_2^s as unified output, which is fed into the BCD prediction head to generate BCD maps.

Based on the proposed DEM, we successfully utilize the semantic information provided by SS to effectively enhance the changed areas of the bi-temporal features. In summary, our DEM further enhances collaboration between BCD and SS.

E. Loss Function

Following the previous studies [11], [16], [41], [42], the cross-entropy loss (CEL) is employed to optimize the classification task of each branch. Specifically, we choose the multiclass CEL as L_{SS} to optimize the SS subtask. For the optimization of the BCD subtask, we adopt the binary CEL as L_{BCD}. The details are shown as follows:

$$L_{\text{SS}} = -\frac{1}{N}\sum_{i=1}^{N} y_i^s \log\left(p_i^s\right) \tag{15}$$

where N represents number of classes, y_i^s and p_i^s denote the ground truth and the predicted probability of the ith class, respectively. L_{BCD} can be expressed as follows:

$$L_{\text{BCD}} = -\frac{1}{N}\sum_{i=1}^{N} y_i^c \log p_i^c + \left(1-y_i^c\right)\log\left(1-p_i^c\right) \tag{16}$$

where y_i^c denotes the change probability in the ith pixel of change map and y_i^c represents the reference value in the ith pixel of ground truth.

Moreover, we also introduce SCLoss [15] as L_{SC} to ensure semantic consistency between the two predicted SS maps, which can be expressed as follows:

$$L_{\text{SC}} = \begin{cases} 1-\cos(x_1, x_2), & y^c = 0 \\ \cos(x_1, x_2), & y^c = 1 \end{cases} \tag{17}$$

where y^c represents the ground truth just as in (16) and x_1 and x_2 are prediction features. The total loss L_{SCD} for the SCD task is calculated as follows:

$$L_{\text{SCD}} = \left(L_{\text{SS}_1} + L_{\text{SS}_2}\right)/2 + L_{\text{BCD}} + L_{\text{SC}}. \tag{18}$$

IV. Experiments

A. Implementation Details

The proposed EC-MCNet is implemented under the deep learning framework of PyTorch, with NVIDIA GeForce RTX 3090 GPU. We train our network 50 epochs on two SCD datasets. The batch size and initial learning rate are 8 and 0.1, respectively. The gradient descent optimization method is stochastic gradient descent with Nesterov momentum. The learning rate is updated at each iteration as: 0.1 $*$ (1-iterations/total_iterations)$^{1.5}$. Moreover, random flipping and rotation are used for data augmentation.

B. Datasets

Two representative SCD datasets are selected in this article to evaluate the proposed EC-MCNet.

1) *SECOND [1]:* The SECOND dataset is composed of 2968 pairs of images, whose resolution ranges from 0.5 to 3 m and size is 512 × 512 pixels. They annotate six categories of ground objects: water, ground, low vegetation, trees, buildings, and sports fields as well as a single "no-change" category. We further split 1/5 of the data (593 pairs of images) as the test set and use the remaining ones for training (2375 pairs of images) following the practice in [15].

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TABLE I

QUANTITATIVE COMPARISONS IN TERMS OF MIOU, SEK, AND F_{SCD} ON TWO SCD DATASETS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Method	Params (M)	Flops (G)	Second	Landsat-SCD
			mIoU (%) / Sek (%) / F_{scd} (%)	mIoU (%) / Sek (%) / F_{scd} (%)
HRSCD-str3(2019) [14]	12.77	42.67	66.80 / 13.23 / 54.30	77.73 / 26.66 / 68.10
HRSCD-str4(2019) [14]	13.71	43.69	72.30 / 20.71 / 60.87	79.59 / 33.86 / 74.35
SSCDl(2022) [15]	23.31	47.44	73.35 / 23.25 / 63.18	82.86 / 44.06 / 81.91
BiSRNet(2022) [15]	23.39	189.91	73.16 / 22.87 / 62.84	81.75 / 41.49 / 80.51
DEFO-MLTSCD(2024) [11]	26.02	401.09	73.71 / 23.77 / 63.64	83.35 / 43.99 / 81.06
TED(2024) [12]	24.19	204.29	73.44 / 23.36 / 63.36	86.50 / 52.98 / 85.84
SCanNet(2024) [12]	27.90	264.95	73.19 / 23.49 / 63.71	86.67 / 54.01 / 86.61
EC-MCNet(ours)	29.12	313.47	73.84 / 24.12 / 64.02	87.47 / 55.70 / 87.04

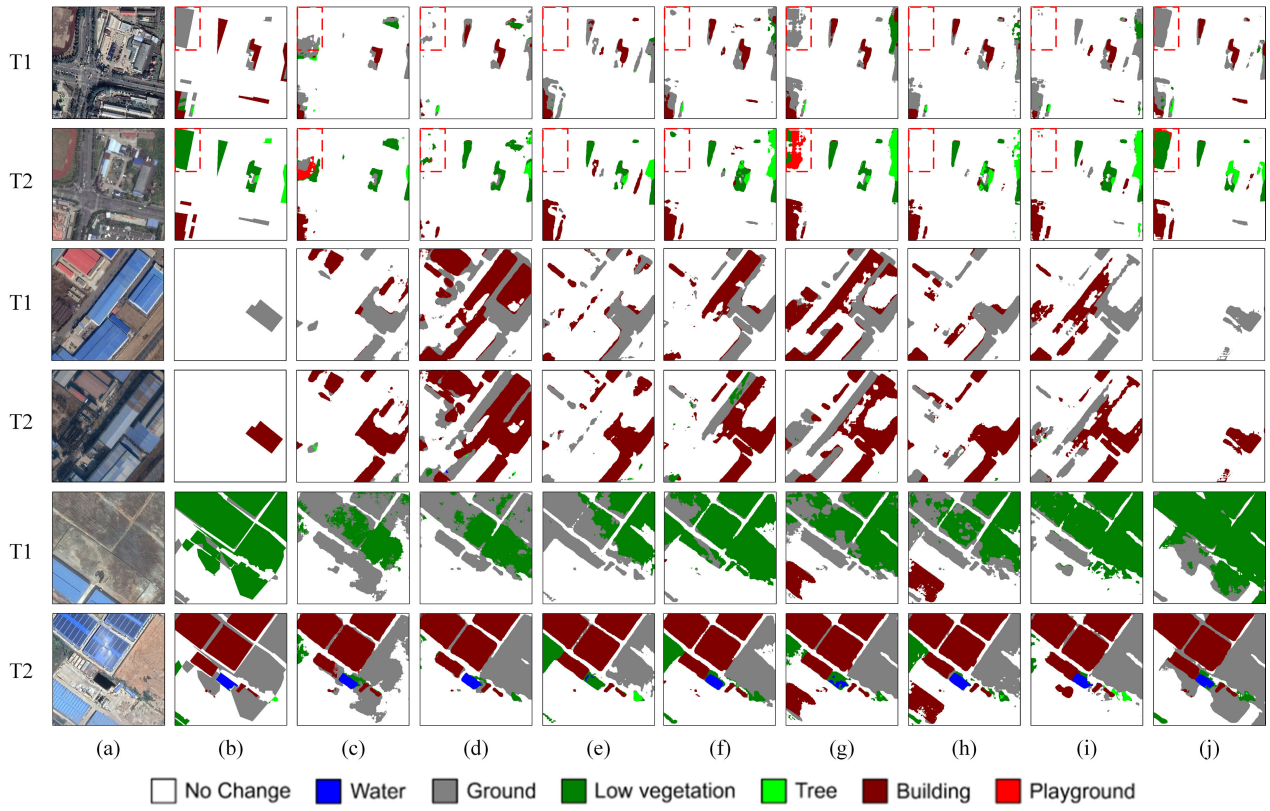


Fig. 3. Visual comparisons of the proposed method and the SOTA methods on the SECOND dataset. Major differences are highlighted with dotted rectangles. (a) Bi-temporal images. (b) Ground truth. (c) HRSCD-str3. (d) HRSCD-str4. (e) SSCDl. (f) BiSRNet. (g) DEFO-MLTSCD. (h) TED. (i) SCanNet. (j) Ours.

2) *Landsat-SCD [2]:* As for the Landsat SCD dataset, there are 8468 pairs of images collected between 1990 and 2020, whose resolution is 30 m and size is 416 × 416 pixels. The dataset is annotated into the no-change category and four categories of ground objects, including farmland, desert, building, and water. We remove the manually enhanced samples to obtain 2385 pairs of completely different images to reduce the data leakage as much as possible while keeping the effective model learning. We further split them into training, validation, and test sets with the numeric ratio of 3:1:1 (i.e., 1455, 485, and 485, respectively).

C. Compared Methods

Seven representative and up-to-date SCD methods are totally selected for comparison to evaluate the performance of the proposed EC-MCNet. Specifically, HRSCD-str3 and HRSCD-str4 [14] are both pioneering studies in SCD using deep learning, where the former adopts a dual-branch structure, and the latter uses a triple-branch structure. SSCDl [15]

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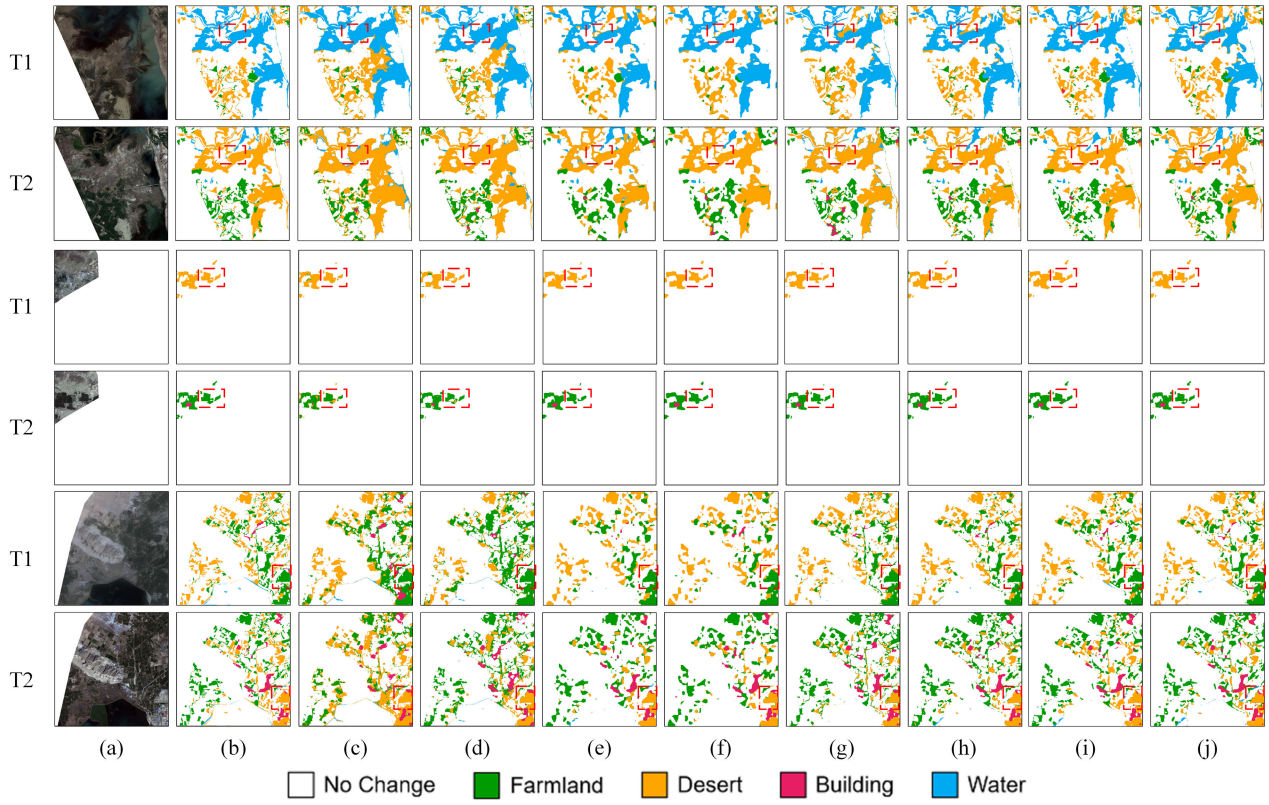


Fig. 4. Visual comparisons of the proposed method and the SOTA methods on the Landsat-SCD dataset. Major differences are highlighted with dotted rectangles. (a) Bi-temporal images. (b) Ground truth. (c) HRSCD-str3. (d) HRSCD-str4. (e) SSCDl. (f) BiSRNet. (g) DEFO-MLTSCD. (h) TED. (i) SCanNet. (j) Ours.

boosts semantic consistency in two SS branches owing to its proposed semantic consistency loss function. The BiSRNet [15] effectively improves the performance of SCD by facilitating interaction between the two temporal branches. DEFO-MLTSCD [11] focuses on the decoder design from a perspective of multitask learning. TED [12] is proposed to detail and refine the spatial representation of features. The SCanNet [12] explores the joint modeling of the spatial–temporal dependency to improve the learning of semantic changes.

D. Evaluation Metrics

To evaluate the performance of diverse methods, we adopt three metrics commonly used in the SCD task, including mean intersection over union (mIoU), separated Kappa (Sek) coefficient introduced in [1], and SCD-targeted $F1$ score (F_{scd}). The specific computation formula is expressed as follows:

$$\begin{aligned} \text{IoU}_0 &= \frac{\text{TN}}{\text{TN}+\text{FP}+\text{pn}} \\ \text{IoU}_1 &= \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \\ \text{mIoU} &= (\text{IoU}_0+\text{IoU}_1)/2 \end{aligned} \tag{19}$$

where TP, FN, FP, and TN represent true positives, false negatives, false positives, and true negatives pixels, respectively,

$$\text{Kappa} = \frac{p_o - p_e}{1 - p_e}$$

$$\text{Sek} = e^{\text{IoU}_1 - 1} * \text{Kappa} \tag{20}$$

where p_o is defined as the sum of the positive sample classifications, which is then divided by the total number of samples per class. And p_e represents the cumulative products of the actual and forecast figures for all respective classes, which are then divided by the square of the total sample size

$$\begin{aligned} P_{\text{scd}} &= \sum_{i=1}^{N} q_{ii} / \sum_{i=1}^{N}\sum_{j=0}^{N} q_{ij} \\ R_{\text{scd}} &= \sum_{i=1}^{N} q_{ii} / \sum_{i=0}^{N}\sum_{j=1}^{N} q_{ij} \\ F_{\text{scd}} &= \frac{2 * P_{\text{scd}} * R_{\text{scd}}}{P_{\text{scd}} + R_{\text{scd}}}. \end{aligned} \tag{21}$$

Fscd is introduced to evaluate the accuracy of SCD results, which focuses on the accuracy obtained to better describe the discrimination of semantic categories in the SCD task. Moreover, we also provide two metrics concerning parameters (Params) and FLOPs. To calculate FLOPs, we take into account calculations involving a pair of 512 × 512-pixel image inputs.

E. Comparison Results With the SOTA Methods

1) Quantitative Evaluation: Table I shows the quantitative evaluation results of the various methods on the SECOND and Landsat-SCD datasets. On the two public datasets,

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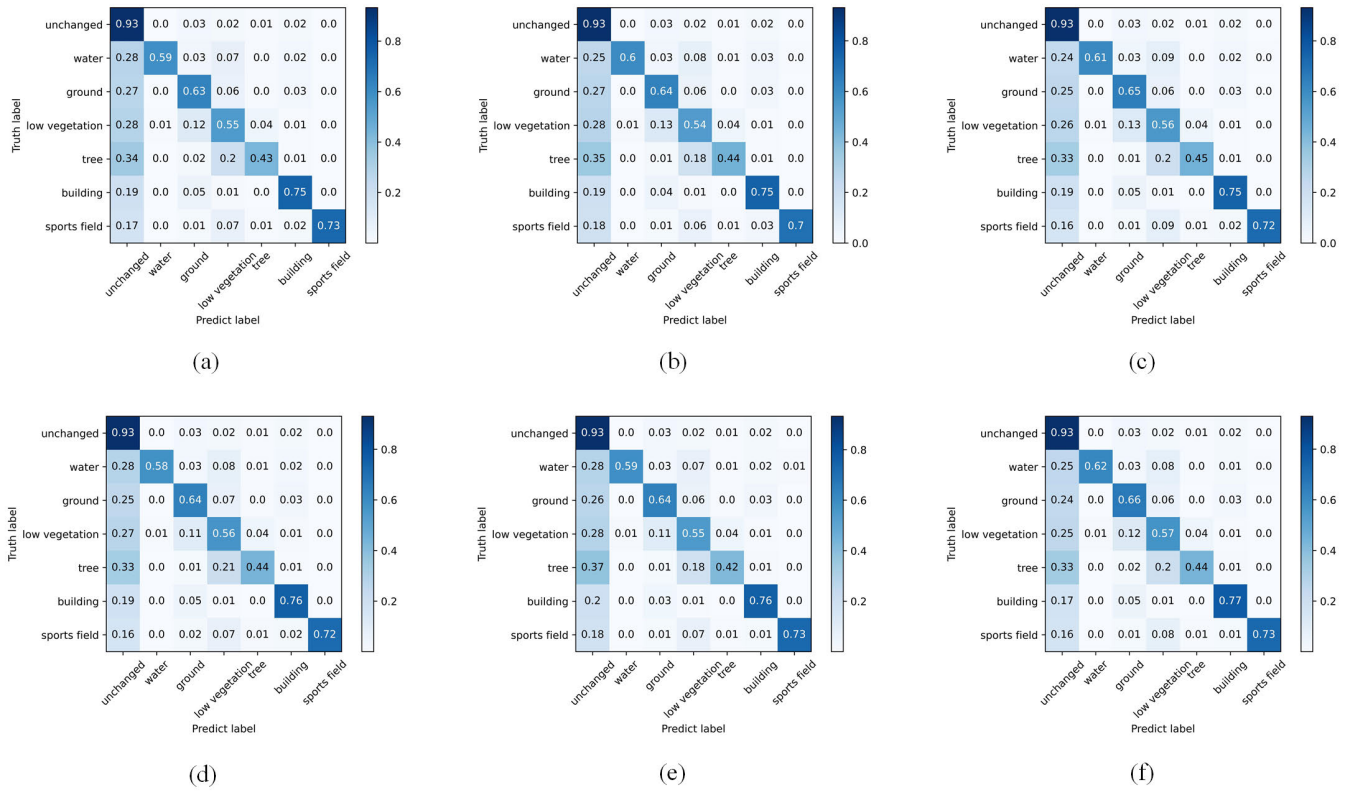


Fig. 5. Confusion matrix visualization results on the SECOND dataset. (a) SSCDl. (b) BiSRNet. (c) DEFO-MLTSCD. (d) TED. (e) SCanNet. (f) Ours.

the proposed EC-MCNet achieves mIoU, Sek, and F_{scd} of 73.84%, 87.47%, and 24.12%, 55.70%, and 64.02%, 87.04%, respectively, having the best performance among these methods. Although we have a larger computational cost and more parameters with respect to most compared methods, we achieve significant performance improvement. Specifically, we improve the mIoU, Sek, and F_{scd} by 0.13%, 0.35%, and 0.31% on the SECOND dataset compared to the second-best method, respectively. Similarly, we improve the mIoU, Sek, and F_{scd} by 0.80%, 1.69%, and 0.43% on the Landsat-SCD dataset compared to the second-best method, respectively. As a result of the above discussion, we argue that the complexity of our method can be acceptable, especially considering the significant performance improvement gains from our method.

2) Visual Results: The visual results of various methods on two SCD datasets are shown in Figs. 3 and 4, major differences are highlighted with dotted rectangles. From the visual results, we can conclude that our method has the following advantages.

1) *Advantage in Accurately Identifying the Changed Areas:* In the first and second rows of Fig. 3, our method can accurately locate the changed areas from ground to low vegetation compared with other methods, which may be because we enhance interclass variation of semantic features as well as establishes the relationship between the semantic feature similarity and changed areas. On the other hand, in the third and fourth rows of Fig. 3, our method can remove pseudo-changes due to the interference of domain styles, which may benefit from efficiently aligning the bi-temporal features by DSRM.
2) *Advantage in Correct and Precise Segmentation of Semantics:* In the fifth and sixth rows of Fig. 3, our method can give correct semantics of low vegetation, ground, and building compared with other methods. Moreover, the results in Fig. 4 also show that the SCD maps of our method are more precise. We believe the reason for this is that we enhance intraclass similarity as well as interclass variation of semantic features and relieve the constraint of SS branch inputs to mine domain-specific features.
3) *Advantage in Consistency of BCD and SS:* In the second, third, fifth, and sixth rows of Fig. 3, the SCD maps of most methods indicate an issue with inconsistency between BCD and SS. Concretely, some methods give the same semantics other than no change to changed areas, which suggests that the BCD and SS are not cooperating effectively, resulting in inconsistencies. Fortunately, our method gives different semantics to changed areas, which indicates that BCD and SS can achieve consistency and can cooperate better in our method.

3) Confusion Matrix: Fig. 5 displays the confusion matrix for all semantic classes of the proposed EC-MCNet and other SCD methods on the SECOND dataset. The diagonal elements of the confusion matrix represent the true positive percentage for all semantic classes. The elements in the first column, except for (0,0), indicate each class's miss detection rate. The remaining elements signify the rate of false detections.

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TABLE II
QUANTITATIVE COMPARISONS OF THE PROPOSED METHOD WITH DIVERSE SETTINGS IN TERMS OF MIOU, SEK, AND F_{SCD} ON TWO SCD DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN RED

Methods	Flops (G)	SEM	DSRM	DEM	Second	Landsat-SCD
					mIoU(%) / Sek(%) / F_{scd}(%)	mIoU(%) / Sek(%) / F_{scd}(%)
#01	226.62	✗	✗	✗	73.21 / 23.19 / 63.25	86.57 / 53.18 / 85.94
#02	250.36	✓	✗	✗	73.40 / 23.48 / 63.49	86.89 / 54.08 / 86.37
#03	270.28	✗	✓	✗	73.32 / 23.14 / 63.25	86.84 / 53.89 / 86.22
#04	246.06	✗	✗	✓	73.54 / 23.49 / 63.36	86.91 / 54.16 / 86.40
#05	294.02	✓	✓	✗	73.69 / 23.81 / 63.66	87.03 / 54.40 / 86.41
#06	289.72	✗	✓	✓	73.64 / 23.53 / 63.53	87.14 / 54.77 / 86.68
#07	269.80	✓	✗	✓	73.68 / 23.88 / 63.77	87.32 / 55.27 / 86.84
EC-MCNet	313.47	✓	✓	✓	73.84 / 24.12 / 64.02	87.47 / 55.70 / 87.04

Excluding the results of (5,5) tested by DEFO-MLTSCD, our method outperforms all other methods, which shows the superiority of our method.

F. Ablation Studies

To validate the effectiveness of the components of the EC-MCNet, we conduct the ablation studies on two SCD datasets. Moreover, the selected low-frequency range in DSRM and the number of groups in the SEM are also discussed.

1) Effectiveness of the SEM: The SEM is designed to enhance interclass variation and intraclass similarity of bi-temporal semantic features. To verify its effectiveness, an ablation experiment (#02 in Table II) is conducted by adding the SEM compared with our baseline (#01 in Table II). The results in Table II show that the better performance can be achieved when we adopted the SEM. Specifically, mIoU values are improved by 0.19% and 0.32% on the two datasets, which suggests that the SEM is effective in our method.

2) Effectiveness of DSRM: The usage of DSRM is a prerequisite for mitigating conflict between BCD and SS, which ensures the inputs of SS branches are relieved of the constraint of mining domain-specific features and bi-temporal domains can be aligned more efficiently. Therefore, we add the DSRM (#03 in Table II) to validate its effectiveness. As shown in Table II, the results of #03 are better than those of #01. Concretely, mIoU values are improved by 0.11% and 0.27% when we adopt DSRM on the two datasets. Meanwhile, we also observe a decrease in the value of Sek on the SECOND dataset. Considering a class imbalance issue in SCD, that is, no-change class is the majority, we argue that DSRM can align bi-temporal features to reduce pseudo-changes that improve mIoU, while it removes some class styles that cause a few unidentified changes or semantic errors that decrease Sek. Fortunately, better performances can be achieved when we combine DSRM with other modules, which may benefit from that SEM can enhance interclass variation and DEM can identify changed areas accurately, subtly tacking the unidentified changes and semantic errors caused by DSRM. To further evaluate the effectiveness of DSRM, we provide visual results of #07 in Table II and EC-MCNet, as shown in Fig. 6.

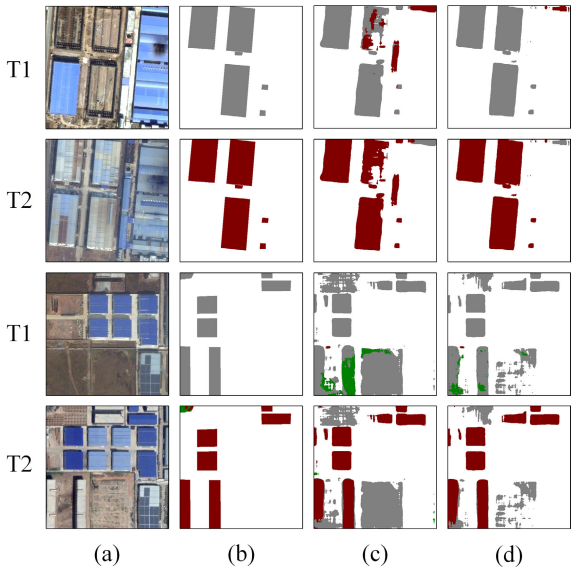


Fig. 6. Visual comparisons regarding DSRM on the SECOND dataset. (a) Bi-temporal images. (b) Ground truth. (c) #07 in Table II. (d) EC-MCNet.

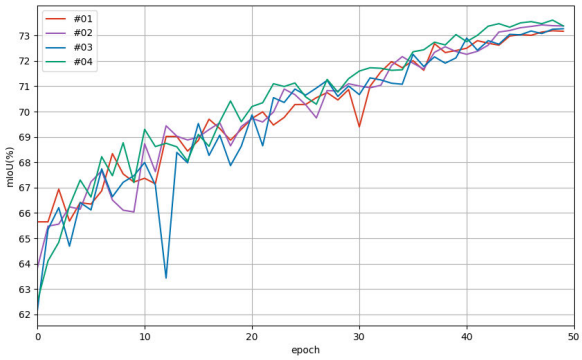


Fig. 7. mIoU values of models for each training epoch. #01, #02, #03, and #04 denote corresponding methods in Table II.

Clearly, our EC-MCNet can reduce pseudo-changes caused by domain styles, exhibiting excellent domain alignment capability. Meanwhile, we also find that more accurate semantics

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TABLE III
ROBUST EVALUATION ON TWO SCD DATASETS, WHICH PRESENTS THE AVERAGE PERFORMANCE ALONG WITH THE STANDARD DEVIATION OVER THREE INDEPENDENT RUNS USING DIFFERENT RANDOM SEEDS

Method	Second	Landsat-SCD
	mIoU (%) / Sek (%) / F_{scd} (%)	mIoU (%) / Sek (%) / F_{scd} (%)
HRSCD-str3(2019) [14]	67.11(±0.25) / 13.71(±0.35) / 54.77(±0.36)	78.02(±0.36) / 25.78(±4.14) / 67.02(±4.15)
HRSCD-str4(2019) [14]	72.37(±0.05) / 20.89(±0.13) / 61.04(±0.13)	79.19(±0.44) / 32.53(±1.68) / 73.37(±1.20)
SSCDl(2022) [15]	73.23(±0.10) / 23.00(±0.21) / 63.01(±0.21)	82.73(±0.23) / 43.78(±0.61) / 81.73(±0.39)
BiSRNet(2022) [15]	73.16(±0.11) / 22.94(±0.21) / 62.97(±0.19)	82.15(±0.45) / 41.67(±0.26) / 81.07(±0.58)
DEFO-MLTSCD(2024) [11]	73.68(±0.03) / 23.66(±0.11) / 63.50(±0.20)	85.42(±1.60) / 49.64(±4.51) / 83.95(±2.18)
TED(2024) [12]	73.36(±0.06) / 23.15(±0.19) / 63.01(±0.31)	86.61(±0.09) / 53.26(±0.25) / 85.98(±0.12)
SCanNet(2024) [12]	73.19(±0.09) / 23.58(±0.14) / 63.79(±0.10)	86.29(±0.35) / 52.98(±1.00) / 86.13(±0.47)
EC-MCNet(ours)	73.83(±0.05) / 24.11(±0.03) / 64.05(±0.03)	87.26(±0.23) / 55.11(±0.73) / 86.76(±0.32)

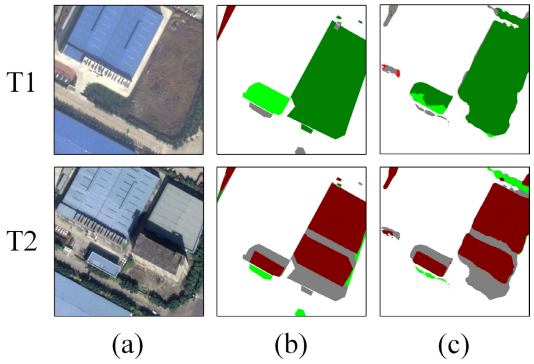


Fig. 8. Failure case of our method on the SECOND dataset. (a) Bi-temporal images. (b) Ground truth. (c) Ours.

in the single domain can be generated by the EC-MCNet, demonstrating better domain-specific class comprehension, which is attributed to the strong mining of domain-specific features. Although DSRM causes a small drop in Sek, most quantitative improvements by it and the visual results can verify its effectiveness.

3) Effectiveness of the DEM: The proposed DEM aims to enhance collaboration between SS and BCD by establishing the relationship between semantic similarity and changed areas. As shown in Table II, mIoU values are improved by 0.33% and 0.34% when we adopt the DEM on the two datasets, which indicates that the DEM effectively improves the performance of SCD.

4) Effectiveness of the Module Combinations: To demonstrate that there is no conflict between our proposed modules, three ablation experiments (#05, #06, and #07 in Table II) are constructed. Fortunately, the results show that better performance can be achieved when we combine our proposed modules and thus suggest there is no conflict between our proposed modules. For instance, the results of #05 are better than those of #02 and #03. Concretely, mIoU of #05 values are improved by 0.29%, 0.14% compared with those of #02 and 0.37%, 0.19% compared with those of #03 on the two datasets. In summary, our EC-MCNet achieves the best performance compared with all variant methods in ablation studies, which shows that not only are the modules valid but also the combination of modules is sensible.

5) Training Stability With Different Modules: As shown in Fig. 7, we present mIoU values of #01, #02, #03, and #04 in Table II versus the number of epochs on the SECOND dataset. We can observe that the introductions of these modules do not affect training stability most of the time and achieve better performances. Meanwhile, we also observe a relatively large fluctuation of #03 (with DSRM) early in training, which may be caused by stochastic gradient descent selecting some samples that are not representative of the domain styles. In summary, we consider the perturbation of training stability by our modules to be acceptable.

6) Robustness Evaluation: As shown in Table III, we present the average performance along with the standard deviation over three independent runs using different random seeds, for evaluating the robustness of our method. Although our method does not have optimal robustness, it is still in a reasonable range. Furthermore, we attempt to analyze the results, which may be caused by the random grouping in SEM and the relatively sophisticated optimization of DSRM. On the other hand, the results further verify the superior capability of our method, whose performance surpasses other methods by large margins.

7) Selected Low-Frequency Range in DSRM: To avoid destroying feature contents while flexibly removing domain styles, we explore the selected low-frequency range in DSRM. In DSRM, the low-frequency components are distributed in the center of the frequency map. Thus, we select a low-frequency component at the center, a 2×2 low-frequency area to the upper left of the center, and a 3×3 area centered on the center. As shown in Table IV, the best performance can be obtained when we select the 2×2 low-frequency area. Meanwhile, the results also indicate selecting more

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TABLE IV
QUANTITATIVE COMPARISONS OF THE EC-MCNET WITH DIVERSE PARAMETER CHOICES. THE RESULTS ARE DESCRIBED IN PERCENTAGE (%), AND THE BEST OF THE RESULTS ARE BOLDED

No.	Setting	Second	Landsat-SCD
		mIoU / Sek / F_{scd}	mIoU / Sek / F_{scd}
Selected low-frequency range in DSRM			
#01	1×1	73.76 / 23.96 / 63.94	87.44 / 55.59 / 86.94
#02	2×2	**73.84 / 24.12 / 64.02**	**87.47 / 55.70 / 87.04**
#03	3×3	73.62 / 23.63 / 63.62	86.92 / 54.10 / 86.31
The number of groups in SEM			
#04	2	73.67 / 23.83 / 63.81	87.14 / 54.79 / 86.70
#05	4	73.80 / 23.97 / 63.86	87.32 / 55.25 / 86.84
#06	8	**73.84 / 24.12 / 64.02**	**87.47 / 55.70 / 87.04**

frequency components degrades model performance while selecting fewer low-frequency components is insufficient for style removal.

8) Number of Groups in the SEM: As a key hyperparameter, the number of groups in the SEM is crucial. Limited by available resources, we only explore the performances when the number of groups is set to 2, 4, and 8 (factors of channels). As shown in Table IV, the best performances on two datasets can be achieved when the number of groups is set to 8, which is close to the number of classes of the SECOND dataset. For the Landsat-SCD dataset, more groups may be more advantageous in the face of its dispersed and complex semantic distributions.

G. Limitations

Although our proposed method has achieved promising results, it still has limitations and failure cases. As shown in Fig. 8, our method confuses trees and low vegetation, causing wrong segmentation results. We believe that the confusion between high-similar classes cannot be fully addressed by grouping strategy, but introducing frequency domain as an additional clue [43] may be a solution. On the other hand, we argue that the stimulation of mining domain-specific features in our method may be insufficient, which can also lead to confusion due to a lack of deeper understanding of domain-specific class differences. In the future, we will explore diverse frequency-based strategies and further stimulation of mining domain-specific features for improving the performance of SCD.

V. CONCLUSION

In this article, we first point out two challenges in existing SCD methods: 1) insufficient collaboration between BCD and SS and 2) the existing conflict between BCD and SS. To address these challenges, we propose EC-MCNet to enhance collaboration and mitigate conflict between SS and BCD. First, we argue that enhancing interclass variation and intraclass similarity across bi-temporal domains is the core of BCD. Similarly, enhancing interclass variation and intraclass similarity is also the core of SS. Therefore, we design the SEM to enhance interclass variation and intraclass similarity of bi-temporal semantic features by grouping and processing the features. Second, we no longer share input but share semantic contents between SS and BCD branches by DSRM, which ensures the inputs of SS branches are relieved of the constraint of mining domain-specific features and bi-temporal domains can be aligned more efficiently. In this way, the conflict between BCD and SS can be efficiently mitigated. Third, inspired by the idea that class-specific representation in SCD is more organized in specific channels compared with BCD, we built DEM to establish the relationship between semantic similarity and changed areas to enhance collaboration between BCD and SS. Finally, we conduct extensive experiments on two public SCD datasets, which demonstrate the performance of the proposed EC-MCNet are the best of the state-of-the-art methods.

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